



Faculty: Information Technology

Academic Year: 2025-2026, Semester: 1

Type of Assessment: Mid Semester Exam

MODULE NAME: Probability and Statistics

Module Code: STAT 8225

Lecturer: Jean Paul NSABIMANA

Date of the Exam: 21st October 2025

Group: Evening (B and G). Exam Duration: 2 Hours

EXAM INSTRUCTIONS:

1. The Paper is made up of **three** questions.
2. Total Marks of the Exam: **30 Marks**.
3. Phones and Electronic Devices are strictly forbidden unless instructed otherwise.
4. Scientific calculators are allowed but not to be shared in the examination room.

Question 1:

Multiple questions

1. If three coins are tossed simultaneously, then the probability of getting two heads together is:
a. $\frac{3}{8}$; b. $\frac{1}{8}$; c. $\frac{5}{8}$; d. $\frac{7}{8}$. (2 Marks)
2. If $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.3$, then $P(A/B)$ is:
a. 0.2; b. 0.3; c. 0.6; d. 0.5. (2 Marks)
3. If events A and B are independent, then $P(A/B)$ is equal to:
a. $P(A)$; b. $P(A)$; c. $1 - P(A)$; d. $P(A) \times P(B)$. (2 Marks)
4. A discrete random variable X has the following probability mass function (pmf):

$$P(X = x) = \frac{x}{10}, \quad x = 1, 2, 3, 4$$

- . What is $P(X \leq 2)$?
a. 0.3; b. 0.5; c. 0.6; d. 0.7. (2 Marks)
5. The probability density function of X is $f(x) = 2x$ for $0 \leq x \leq 1$. What is $P(X \leq 0.5)$?
a. 0.5; b. 1; c. 0.75; d. 0.25. (2 Marks)

Question 2 (10 Marks)

1. Find the value of n in

$$\binom{n}{4} = 13 \binom{n}{2}. \quad (2 \text{ Marks})$$

2. What is the probability of getting the sum as a prime number if two dice are thrown?
(2 Marks)
3. Consider the discrete random variable X with the following distribution

X	1	2	3	4
$P(X = x)$	0.5	0.15	0.1	0.25

Calculate the following:

- (a) the expected value of X , $E[X]$. (2 Mark)
- (b) the expected value of $2X + 3$, $E[3X + 2]$ (1 Mark)
- (c) the variance, $Var[X]$ (2 Marks)
- (d) the standard deviation, σ_X . (1 Mark)

Question 3 (10 Marks)

1. In how many ways can we select 6 people out of 10, of which a particular person is not included? (2 Marks)
2. In a box, there are 8 orange, 7 white, and 6 blue balls. If a ball is picked up randomly, what is the probability that it is neither orange nor blue? (2 Marks)
3. A random variable X has a probability density function

$$f(x) = \begin{cases} kx(1-x) & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the value of k , expectation, variance and standard deviation. (6 Marks)

Success and God Be with You!!